Computer Networks

Lecture 28 Network Layer

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Content

- **Path Vector Routing**

Introduction

Distance vector and link state routing are intradomain protocols used within an autonomous system, but they are not suitable for interdomain routing due to scalability issues.

- Distance vector routing is subject to instability if there are more than a few hops in the domain of operation.
- Link state routing needs a huge amount of resources to calculate routing tables. It also creates heavy traffic because of flooding.

There is a need for a third routing protocol which we call ***path vector routing.***

Path Vector Routing

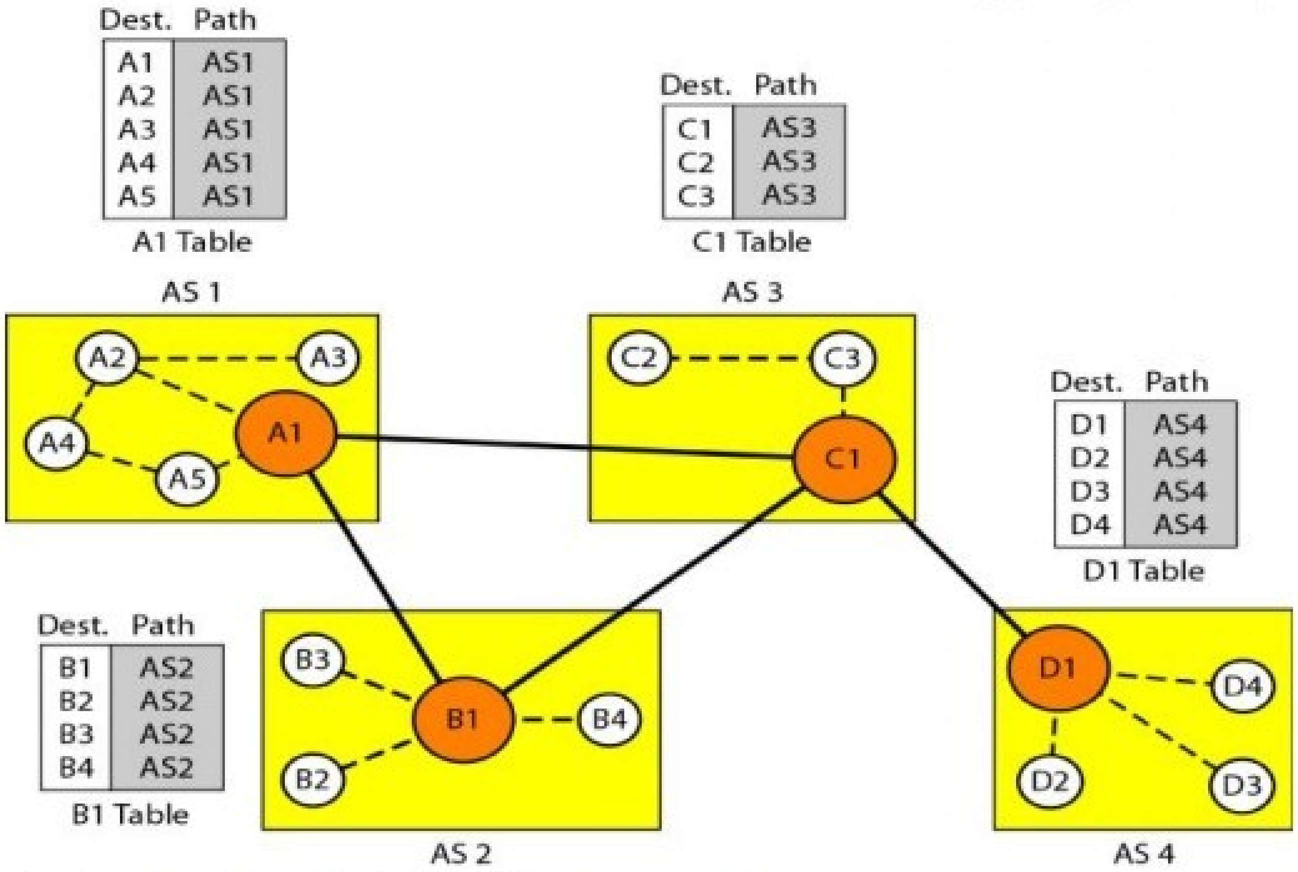
Path vector routing is similar to distance vector routing, but instead of sharing distances (metrics), it shares the entire path to reach a destination. Communication happens between special nodes called speaker nodes, each representing an autonomous system.

- Each AS has a speaker node (represents the whole network)
- Speaker nodes exchange routing information with neighboring ASs
- Advertises complete path (list of ASs), not just distance/metric
- Helps in avoiding routing loops

Path Vector Routing

Initialization

At the beginning, each speaker node can know only the reachability of nodes inside its autonomous system.

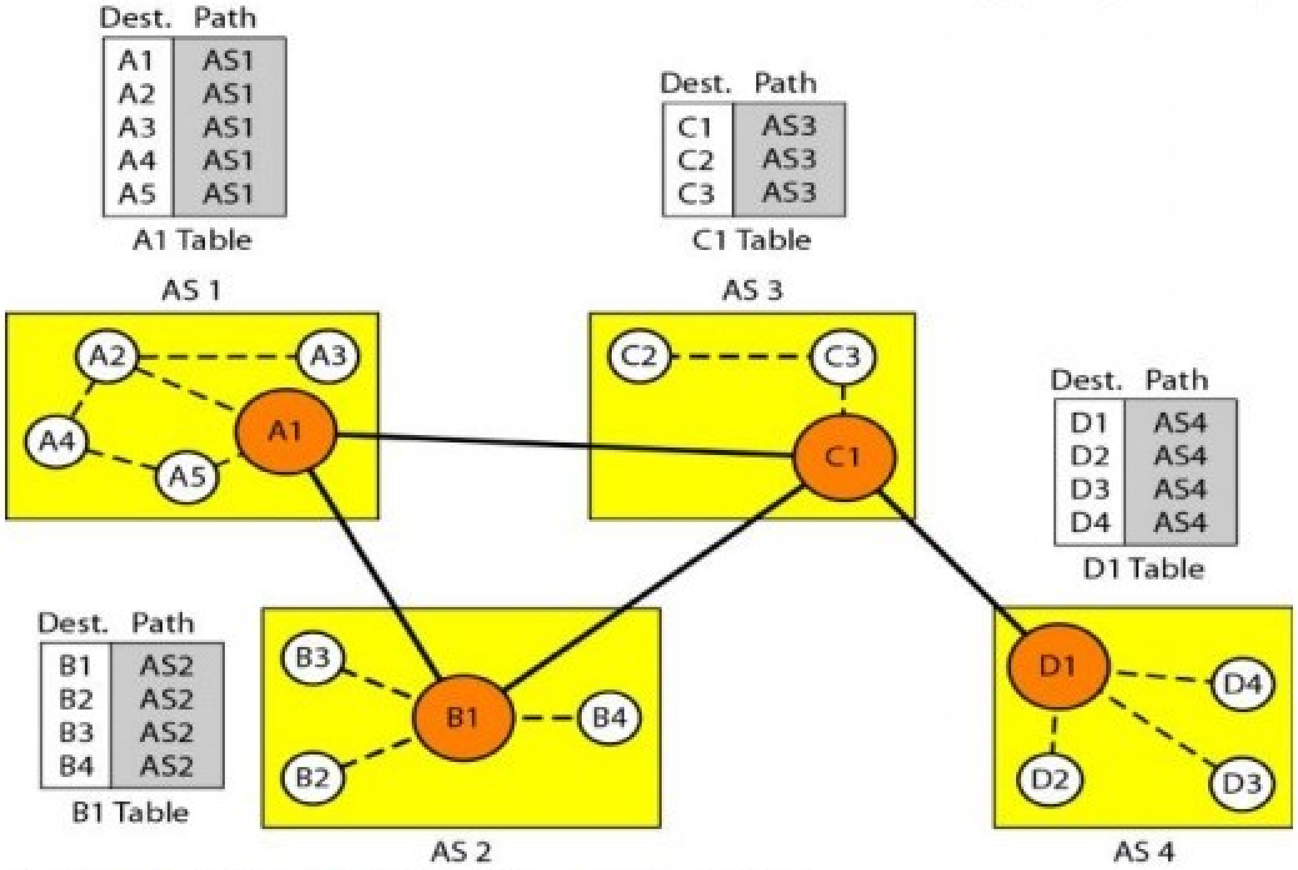


Path Vector Routing

Advertise Table

Node A_1 is the speaker node for AS_1, B_1 for AS_2, C_1 for AS_3, and D_1 for AS_4.

Node A_1 creates an initial table that shows A_1 to A_5 are located in AS_1 and can be reached through it. Node B_1 advertises that B_1 to B_4 are located in AS_2 and can be reached through B_1 . And so on.



A speaker in an autonomous system shares its table with immediate neighbors.

Path Vector Routing

Update Table

When a speaker node receives a two-column table from a neighbor, it updates its own table by adding the nodes that are not in its routing table and adding its own autonomous system and the autonomous system that sent the table.

After a while each speaker has a table and knows how to reach each node in other ASs.

Dest.	Path
A1	AS1
...	
A5	AS1
B1	AS1-AS2
...	...
B4	AS1-AS2
C1	AS1-AS3
...	...
C3	AS1-AS3
D1	AS1-AS2-AS4
...	...
D4	AS1-AS2-AS4

A1 Table

Dest.	Path
A1	AS2-AS1
...	
A5	AS2-AS1
B1	AS2
...	...
B4	AS2
C1	AS2-AS3
...	...
C3	AS2-AS3
D1	AS2-AS3-AS4
...	...
D4	AS2-AS3-AS4

B1 Table

Dest.	Path
A1	AS3-AS1
...	
A5	AS3-AS1
B1	AS3-AS2
...	...
B4	AS3-AS2
C1	AS3
...	...
C3	AS3
D1	AS3-AS4
...	...
D4	AS3-AS4

C1 Table

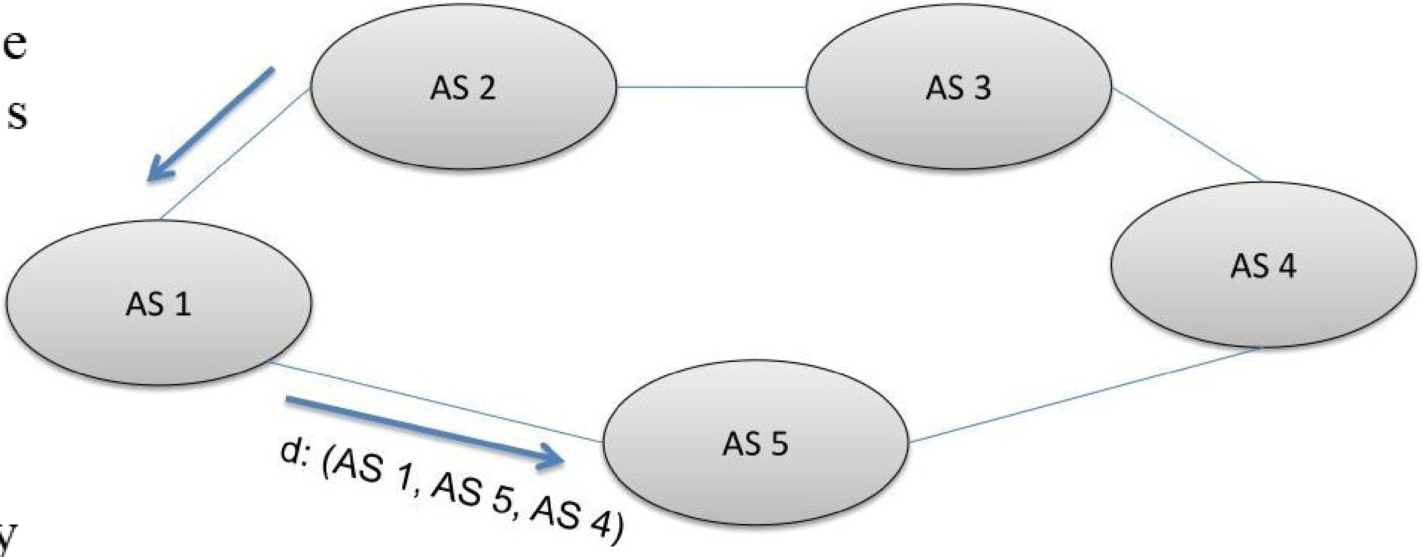
Dest.	Path
A1	AS4-AS3-AS1
...	
A5	AS4-AS3-AS1
B1	AS4-AS3-AS2
...	...
B4	AS4-AS3-AS2
C1	AS4-AS3
...	...
C3	AS4-AS3
D1	AS4
...	...
D4	AS4

D1 Table

Path Vector Routing

Loop prevention.

- Each route advertisement carries the entire path (list of Autonomous Systems)
- When a router receives a route:
  - It checks the $AS - path$ list
  - If its own AS appears in the path $\rightarrow$ loop detected
  - The route is discarded immediately
- This prevents routing loops, unlike distance vector routing



Path Vector Routing

Policy routing

When a router receives a message, it can check the path. If one of the autonomous systems listed in the path is against its policy, it can ignore that path and that destination.

Optimum path

The optimum path is the path that fits the organization.

Questions

Q1 A network has nodes A, B, C, D with link costs:

- $A–B = 2$, $A–C = 5$
- $B–C = 1$, $B–D = 4$
- $C–D = 2$

Using Distance Vector routing, find the routing table at node A after convergence.

Questions

Q2 **Network:** $AS1 — AS2 — AS3 — AS4$ (Destination D connected to AS4)

- AS4 advertises to all: [AS4] → D
- Later On AS2 receives an update from AS1: $[AS1,\ AS2,\ AS3,\ AS4]$

1. What path does AS1 store for D?
2. Will AS2 accept this update?
3. Compare with distance vector—what would happen?

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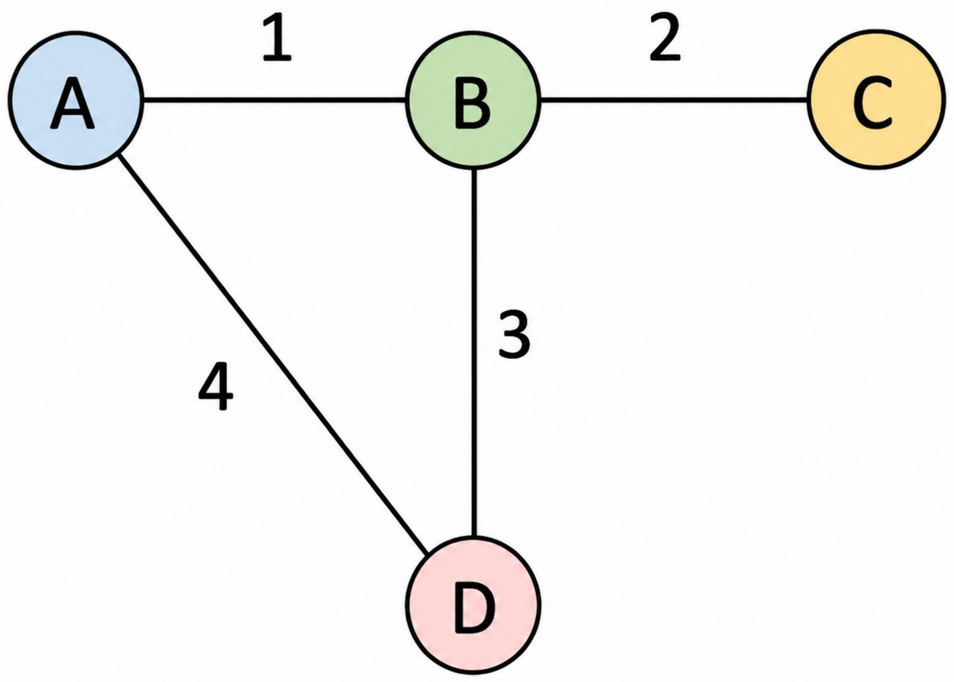
Questions

1. Compute shortest path from A using Dijkstra’s Algorithm
2. Simulate distance vector update for 2 iterations
3. Compare results



Question

An inter-domain network consists of four autonomous systems: AS1, AS2, AS3, and AS4, where destination D is directly connected to AS4.

The network connections and link costs are as follows:

- Cost between AS1 and AS2 = 2
- Cost between AS2 and AS4 = 2
- Cost between AS1 and AS3 = 1
- Cost between AS3 and AS4 = 5

Routing advertisements are given below:

- AS4 advertises destination D as: [AS4]
- AS2 advertises: [AS2, AS4]
- AS3 advertises: [AS3, AS4]

Using **link state routing** (shortest path approach), determine the optimal path from **AS1 to destination D**. Show all cost calculations.